Blood Cell Classification Using Deep Learning Methods

Marina Nouchou
Department of Digital Systems
University of Piraeus
Piraeus, Greece
me2423@webmail.unipi.gr

Panagiotis Athanasopoulos
Department of Digital Systems
University of Piraeus
Piraeus, Greece
me2402@webmail.unipi.gr

***Abstract*—This study focuses on the automatic classification of white blood cells (WBCs) through deep learning, using the Blood Cell Images Dataset. Several architectures were evaluated, with particular emphasis on the ResNet18 model, which achieved the best performance with 99.95% accuracy. In addition, interpretability techniques such as Grad-CAM++ and the Reliability Diagram were applied, enhancing the model's reliability and its potential use in clinical applications.**

The performance of ResNet18 was compared with other methods from the literature, such as EfficientNetB3, DenseNet121, and MobileNet, demonstrating superiority in both accuracy and generalization. Optimization techniques such as normalization, dropout, and data augmentation were also used, while the comparison was conducted based on multiple metrics and confusion matrices.

Overall, the results indicate that the appropriate choice of architecture, combined with interpretability and data augmentation methods, can lead to accurate and reliable diagnosis through automatic cell classification. For future research, integration of the model into clinical diagnostic systems and its application to larger populations and different cell types are proposed.

***Index Terms*—Blood cell classification, Deep learning, CNN, VGG16, ResNet18, Grad-CAM++**

I. Introduction

Automatic blood cell classification is a critical process for assisting pathologists in accurate diagnosis in hematology. The process involves categorizing microscopic images of white blood cells into their corresponding classes. The use of deep learning techniques, and specifically convolutional neural networks (CNNs), has shown significant progress in this field by exploiting their ability to automatically extract and learn features from image data.

The present study is based on the **Blood Cell Images Dataset**, which is available on the Kaggle platform [1]. The dataset includes four categories of white blood cells: Eosinophil, Lymphocyte, Monocyte, and Neutrophil. The objective is to develop and compare three deep learning architectures—CNN, VGG16, and ResNet18—for the classification of these cells, evaluating their performance using metrics such as accuracy, recall, and F1-score.

The evaluation is carried out using a methodology that includes image preprocessing, data augmentation, as well as interpretability techniques such as Grad-CAM++ maps, in order to highlight the image regions that influence the predictions of the models.

The contribution of this study lies in the detailed comparison of the three models on a balanced and augmented dataset, providing useful indications for selecting an appropriate architecture for hematology applications.

A number of recent research studies have used the same dataset to develop and evaluate white blood cell classification models with impressive results.

In the work of Kumar and Nelson [2], the EfficientNetB3 architecture was used with fine-tuning of the final layers and extensive data augmentation, achieving an overall accuracy of 99.63%.

Asghar et al. [3] developed a custom CNN model inspired by VGG and Inception structures, enhanced with batch normalization, dropout, and data augmentation techniques such as rotations and random translations, reaching an accuracy of 99.57%.

Nahzat et al. [4] designed a lightweight CNN architecture using the RMSprop optimizer and no external pretrained weights. Despite the simplicity of the approach, their model achieved an accuracy of 99.5%.

Girdhar et al. [5] followed a similar approach using a simple CNN and basic data preprocessing, including grayscale conversion and normalization, reporting an accuracy of 98.55%.

Rehman et al. [6] used a pretrained VGG16 model with fine-tuning and dropout in the fully connected layers, as well as cross-entropy loss for multiclass classification, achieving noteworthy performance.

Transfer learning is widely used in many studies. Biswal et al. [7] applied Inception-ResNet-v2 in a fully automated procedure, without manual feature extraction or traditional image preprocessing, reporting an accuracy of 99%.

Sevinç et al. [8] incorporated a genetic algorithm (GA) for hyperparameter tuning, such as the number of filters and batch size, in combination with D-CNN, resulting in accuracy above 93%.

Davamani et al. [9] conducted a comparative study of six different models, including VGG16, MobileNet, EfficientNetB0, and ResNet, on a balanced dataset. VGG16 and MobileNet emerged as the best models, with accuracy reaching up to 100% for individual classes.

In the work of Sharma et al. [10], DenseNet121 was applied with normalization techniques such as batch normalization and data augmentation, including random flipping and zooming, reaching an overall accuracy of 98.84%.

Finally, Yıldırım and Cinar [11] combined traditional image filters, including Gaussian and median filters, with CNNs to reduce noise and improve the visual clarity of the data. The use of these filters improved the quality of model training and led to high classification performance.

The variety of methods, architectures, and preprocessing techniques found in the literature demonstrates the widespread acceptance and reliability of the Kaggle Blood Cell Dataset as a benchmark, while also highlighting the need for a targeted comparison of models under common experimental conditions.

II. Dataset and Preprocessing

The **Blood Cell Images** dataset from Kaggle was used to train and evaluate the system. It contains four classes of white blood cells: Eosinophil, Lymphocyte, Monocyte, and Neutrophil. The images were preprocessed by resizing them to 224×224 pixels and normalizing the RGB values using ImageNet statistics.

To improve generalization and increase the size of the training set, extensive **data augmentation** was applied. This included random horizontal flips, rotations, translations, scaling, changes in brightness, contrast, and saturation, as well as Random Erasing. In addition, specialized transformations with different intensity levels and parameters were designed for underrepresented classes such as Eosinophil and Neutrophil, so that additional examples could be generated through oversampling.

The final dataset was divided into a training set (80%) and a validation set (20%) using stratified sampling in order to preserve the proportions among the classes.

For the loss function, Focal Loss was selected instead of the conventional Cross-Entropy loss in order to address class imbalance more effectively. Focal Loss reduces the contribution of “easy” examples and increases the model’s focus on “difficult” cases. It is defined as:

$$\mathcal{L}_{\text{focal}} = -\alpha_t (1 - p_t)^{\gamma} \log(p_t)$$

where p_t is the probability assigned to the correct class, γ is the focusing hyperparameter, typically set to 2, and α_t is the weighting factor for each class, which was used according to the frequency of the samples. This approach is based on the work of Lin et al. [12], where it proved particularly effective in problems with substantial class imbalance.

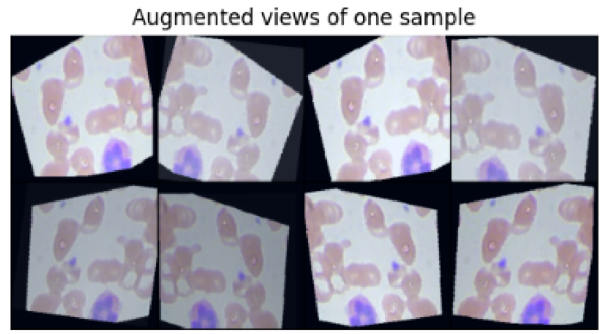

Figure 1: Augmented views of a sample blood cell image.

III. Models and Methodology

A. Model Architectures

Within the scope of this study, three different deep learning models were implemented and evaluated for white blood cell classification, with the aim of comparatively analyzing their performance.

1. Custom Convolutional Neural Network (CNN): The CNN model was designed from scratch in order to provide a lightweight and fast classifier adapted to the size and nature of the dataset. It consists of four convolutional blocks with an increasing number of filters (32, 64, 128, 256), each accompanied by Batch Normalization and ReLU activation. The use of MaxPooling layers (2×2) after each block reduces the spatial dimensions and supports the extraction of localized features.

At the end of the network, two fully connected layers transform the spatial features into final class predictions. To address overfitting, Dropout with a rate of 0.5 was applied to the fully connected layers. CNNs of this type have shown impressive performance when combined with data augmentation and specialized loss functions such as Focal Loss [3], [5], which focuses on difficult-to-classify examples and can improve accuracy in imbalanced datasets.

2. VGG16 (Transfer Learning with Fine-Tuning): The VGG16 architecture is one of the most fundamental and effective CNNs, originally proposed by Simonyan and Zisserman [13]. Its structure is based on successive convolutional layers with small 3×3 filters followed by MaxPooling, enabling gradual and efficient feature extraction. In this study, the ImageNet-pretrained version of the model was used, while fine-tuning was applied to the last five convolutional layers so that the network could adapt to the cellular image domain.

The final classification layer was replaced by a new fully connected layer with four outputs corresponding to the Eosinophil, Lymphocyte, Monocyte, and Neutrophil classes. Transfer learning makes it possible to reuse general features learned from large datasets such as ImageNet and specialize them through fine-tuning on domain-specific medical data. VGG16 has been widely used in similar white blood cell classification applications with high accuracy [6], [9].

3. ResNet18 (Residual Learning): The ResNet (Residual Network) architecture was introduced by He et al. [14], introducing the concept of residual connections, also known as skip connections, which allow information to be transmitted directly between non-adjacent layers. This addresses the vanishing-gradient problem and enables the training of much deeper networks.

ResNet18 contains 18 layers with residual blocks and was used as a pretrained model with fine-tuning of the final layers. As with VGG16, the final fully connected layer was modified to contain four outputs. The initial layers were kept frozen in order to preserve general ImageNet features. ResNet18 is known for providing a good balance between computational efficiency and performance in medical image classification, and it has also been successfully applied in other studies using the same dataset [15], [7].

The selection of these three models allows the evaluation of custom, pretrained, and state-of-the-art architectures within a common experimental framework.

In summary, the selected architectures cover a spectrum ranging from custom, simple CNNs to advanced pretrained structures using residual learning and transfer learning, enabling comparative evaluation under a common framework and dataset.

B. Loss Function

To address class imbalance and focus on difficult examples, Focal Loss was used instead of the standard CrossEntropyLoss. Focal Loss adds a weighting factor that emphasizes misclassified samples and is defined as:

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$$

where $\gamma = 2$ and the α_t weights were manually defined based on the class distribution.

C. Data Augmentation and Resampling

Extensive data augmentation techniques were applied, including horizontal flipping, rotation, affine transformations, and color distortions. For minority classes, such as EOSINOPHIL and NEUTROPHIL, resampling was performed by multiplying samples and applying aggressive augmentation, creating a more balanced distribution in the training set.

D. Evaluation Metrics

The models were evaluated based on accuracy, per-class precision, recall, F1-score, ROC and Precision-Recall curves, and confusion matrices. Feature embeddings were visualized through t-SNE, while calibration metrics such as reliability diagrams, the Brier score, and Expected Calibration Error (ECE) were also reported.

E. Data Augmentation and Oversampling

Extensive data augmentation was applied to improve model generalization and increase the diversity of the training data. The techniques used included horizontal flips, rotations, affine transformations, as well as color modifications through variations in brightness, contrast, and saturation.

To address class imbalance—particularly for underrepresented classes such as *Eosinophil* and *Neutrophil*—oversampling was applied. Specifically, minority-class samples were multiplied through aggressive augmentation transformations in order to create a more balanced training set.

F. Training Details

Each model was trained for up to 20 epochs using the *Adam* optimizer, which provides rapid convergence through an adaptive learning rate for each parameter. An *early stopping* mechanism based on the validation loss was applied to reduce the risk of overfitting.

The division of the data into training and validation sets was performed using *stratified sampling*, preserving the original class proportions.

G. Evaluation Metrics

The models were evaluated using several performance metrics, including accuracy, per-class precision, recall, and F1-score. In addition, ROC and Precision-Recall (PR) curves, as well as confusion matrices, were used to analyze classification errors.

For visual interpretation of the features learned by each model, t-SNE was applied to the feature embeddings. Finally, calibration metrics such as reliability diagrams, the Brier score, and Expected Calibration Error (ECE) were examined in order to evaluate the agreement between predicted probabilities and the true distribution.

H. Interpretability through Grad-CAM++

Grad-CAM++ was used to interpret the decisions of the models by visualizing the image regions that most strongly influenced the final prediction.

For each cell category, one representative image was selected and passed through the model, after which the corresponding heatmap was generated. This heatmap was overlaid on the original image in order to highlight the regions that activated the network most strongly for the specific prediction.

IV. Results - CNN Model

A. Performance Metrics

The custom CNN model was evaluated using several performance metrics, revealing its excellent ability to classify white blood cells.

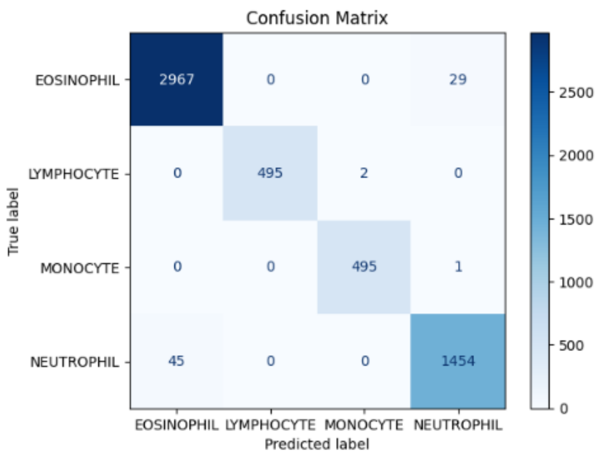

Figure 2: Confusion Matrix for the CNN model.

The confusion matrix shows very low error rates among the classes. The model correctly classified almost all samples, with only a small number of incorrect predictions between Neutrophil and Eosinophil.

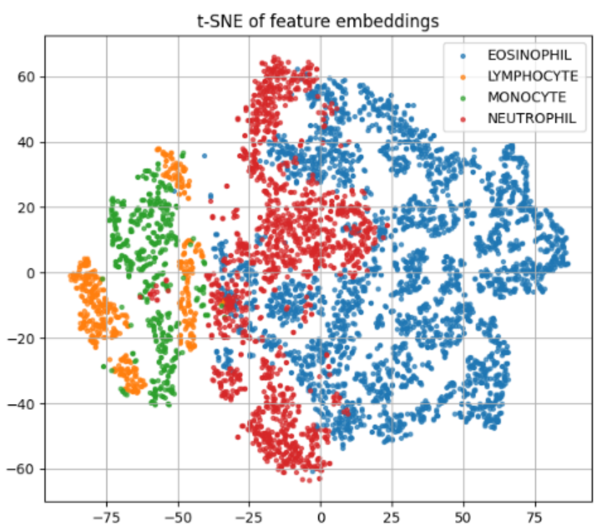

Figure 3: t-SNE feature visualization for the CNN model.

The t-SNE analysis shows clear separability among the four classes, confirming that the model extracts distinct and coherent features for each category.

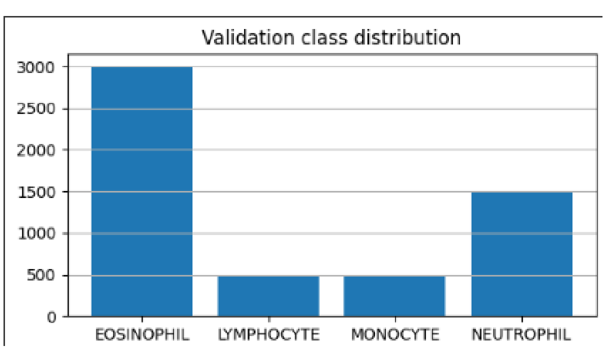

Figure 4: Sample distribution in the validation set.

The sample distribution in the validation set shows overrepresentation of the Eosinophil class. Nevertheless, the model maintains high performance even for less strongly represented categories.

Table I: Classification Report for the CNN Model

Class	Precision	Recall	F1-score	Support
EOSINOPHIL	0.99	0.99	0.99	2996
LYMPHOCYTE	1.00	1.00	1.00	497
MONOCYTE	1.00	1.00	1.00	496
NEUTROPHIL	0.98	0.97	0.97	1499
Accuracy		0.99 (5488 samples)		
Macro avg	0.99	0.99	0.99	5488
Weighted avg	0.99	0.99	0.99	5488

The overall accuracy of the model reaches 99%, with all F1-scores ranging between 0.97 and 1.00, confirming the reliable performance of the CNN for every class. Particularly impressive is the performance for the Lymphocyte and Monocyte categories, where classification is nearly perfect.

B. Training Curves

The training curve of the CNN model shows a rapid decrease in the loss value for both the training and validation sets during the first epochs, indicating successful feature learning. From the 7th epoch onward, the loss stabilizes with small fluctuations and without significant signs of overfitting, while validation accuracy remains above 97%, reaching a peak of approximately 98.5%.

In addition, the reliability diagram shows good agreement between predicted probabilities and actual accuracy, particularly at high confidence levels. This indicates that the model is well calibrated and that its predictions are not only accurate but also reliable.

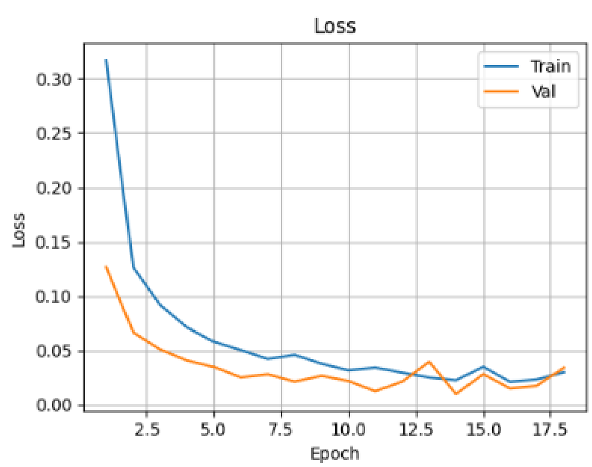

Figure 5: Training and validation loss for the CNN model.

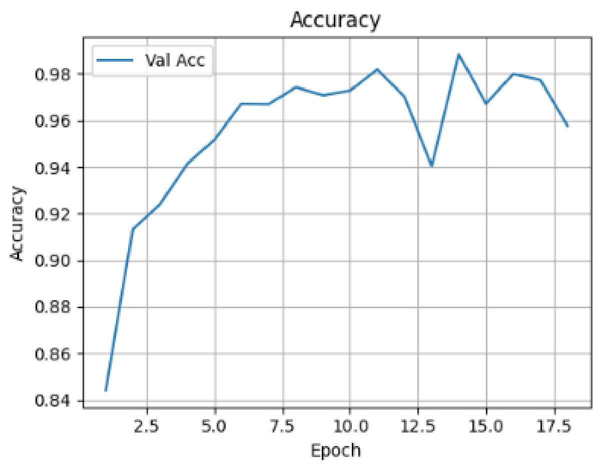

Figure 6: Evolution of validation accuracy for the CNN model.

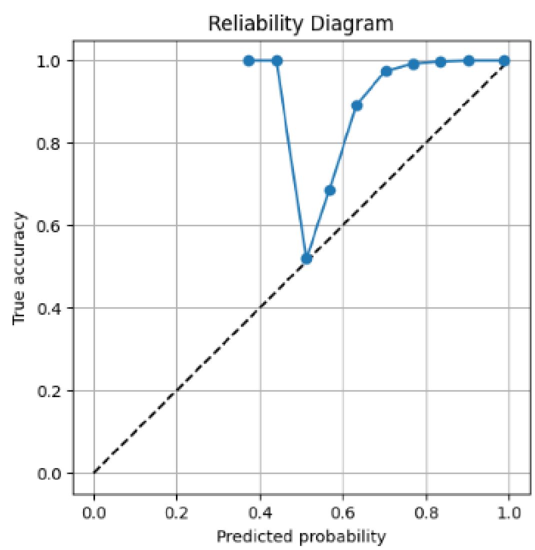

Figure 7: Reliability Diagram for the CNN predictions.

C. Precision-Recall and ROC Curves

Precision-Recall (PR) and ROC curves are used to evaluate the classification ability of the model for each class separately. PR curves illustrate the relationship between precision and recall and are particularly useful in cases of imbalanced data, while ROC curves depict the true positive rate against the false positive rate. In this study, both types of curves show excellent performance, with AUC values close to 1.00, indicating the high accuracy of the CNN model in recognizing all white blood cell types.

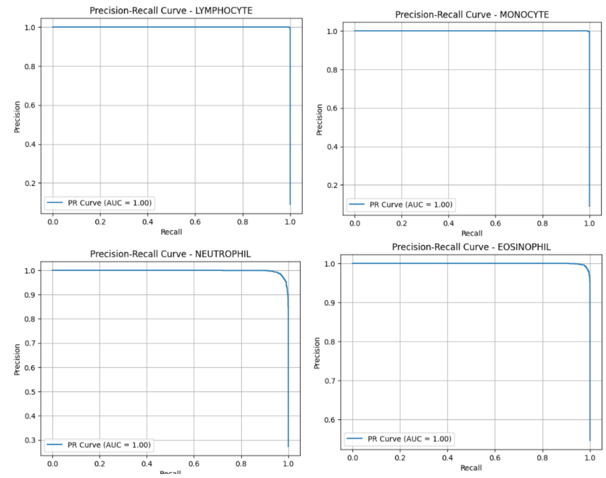

Figure 8: Precision-Recall curves per class for the CNN model.

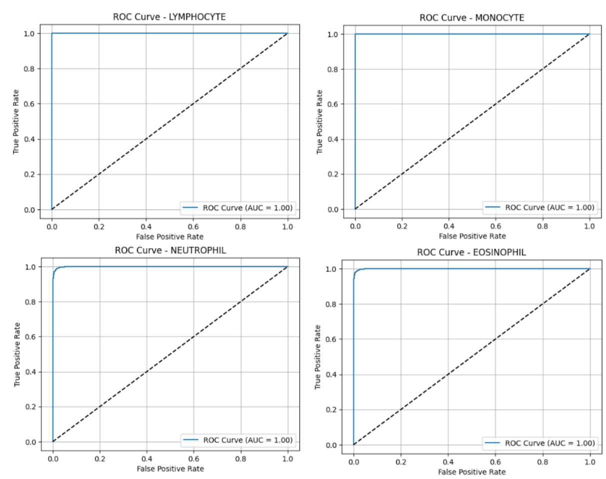

Figure 9: ROC curves per class for the CNN model.

D. Grad-CAM++ Visualization

The Grad-CAM++ visualizations show the image regions that contributed most strongly to the model's predictions for each cell class. Warm regions (red/yellow) indicate high importance, while cooler regions (blue) indicate lower contribution. For each white blood cell type, the model appears to focus on morphologically distinctive characteristics:

- **Eosinophil:** Regions with large, strongly granular cytoplasmic structures are identified, which are characteristic of eosinophils.
- **Lymphocyte:** The model's attention is concentrated on the round, compact nucleus with limited cytoplasm, a characteristic feature of lymphocytes.
- **Monocyte:** The warm regions overlap the asymmetric, kidney-shaped nucleus as well as the broad cytoplasm that characterizes monocytes.
- **Neutrophil:** The model focuses on the multilobed nucleus and the presence of neutral granules in the cytoplasm, which are typical characteristics of neutrophils.

These visualizations support the view that the model does not rely on random regions of the image but instead recognizes structural features associated with the biological identity of the cells, strengthening the reliability of the modeling approach.

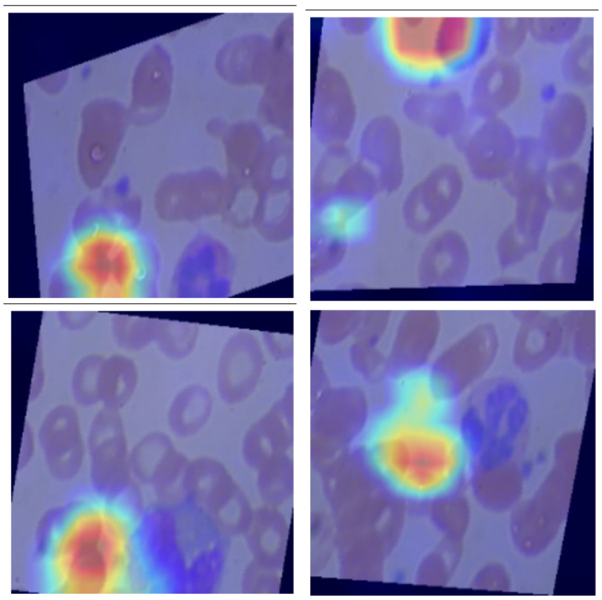

Figure 10: Grad-CAM++ visualizations for each white blood cell category, aimed at interpreting the regions of interest involved in the model predictions.

V. Results - VGG16 Model

A. Confusion Matrix and Overall Metrics

The confusion matrix of the VGG16 model shows very high rates of correct classification for all white blood cell categories. Most samples were classified correctly, with only a few errors, mainly between Eosinophil and Neutrophil. The overall accuracy of the model reaches 98%, while the individual *precision*, *recall*, and *F1-score* metrics exceed 97%, confirming the reliability of the model.

In addition, the detailed per-class classification report shows very high *precision* and *recall* values for Monocyte and Lymphocyte cells, while slightly lower performance is observed for Eosinophils. Overall, the VGG16 model achieves excellent performance in automatic white blood cell classification, with a **macro average F1-score of 97.9%** and a **weighted average F1-score of 98.0%**.

Table II: Detailed classification report for the VGG16 model.

Class	Precision	Recall	F1-score	Support
EOSINOPHIL	0.9807	0.9158	0.9472	499
LYMPHOCYTE	1.0000	0.9859	0.9929	497
MONOCYTE	0.9900	0.9960	0.9930	496
NEUTROPHIL	0.9714	0.9953	0.9832	1499
Accuracy			0.9806	2991
Macro avg	0.9855	0.9733	0.9791	2991
Weighted avg	0.9808	0.9806	0.9804	2991

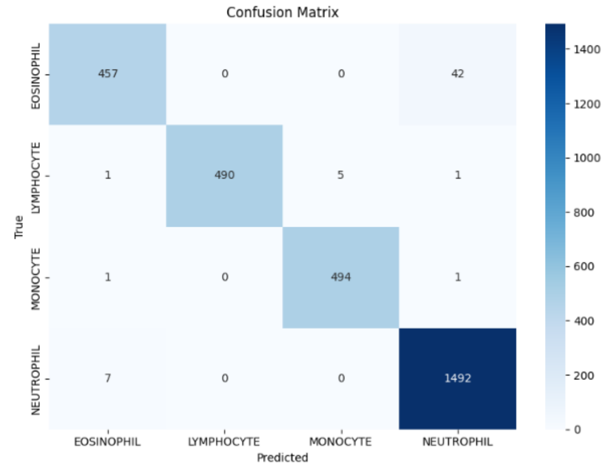

Figure 11: Confusion matrix for the VGG16 model.

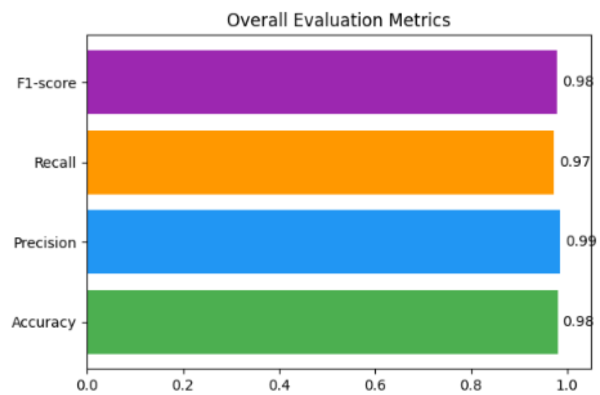

Figure 12: Overall evaluation metrics for the VGG16 model (Accuracy, Precision, Recall, F1-score).

B. Training Curves and Per-Class Evaluation

During training of the **VGG16** model, a steady decrease in the loss value is observed for both the training and validation datasets, as shown in the Loss Curve (Figure 13). This behavior indicates effective model learning without evidence of overtraining.

Validation accuracy (Figure 14) remains high and stable throughout training, demonstrating the model's ability to generalize well to unseen data.

The per-class evaluation (Figure 15) shows excellent performance for all white blood cell categories. The *precision*, *recall*, and *F1-score* values are very high in all cases, with small deviations such as the slightly lower *recall* for the **EOSINOPHIL** category. Overall, the VGG16 model provides reliable and accurate classification across all classes.

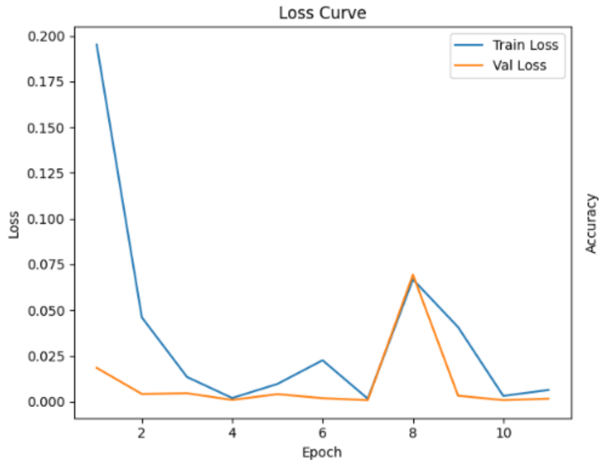

Figure 13: Training and validation loss curve for the VGG16 model.

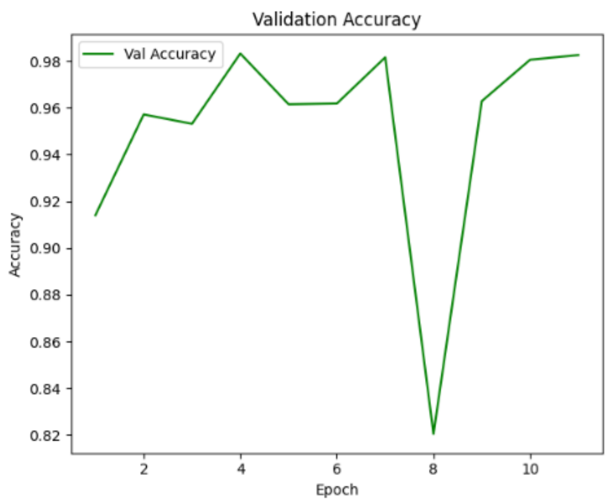

Figure 14: Evolution of validation accuracy for the VGG16 model.

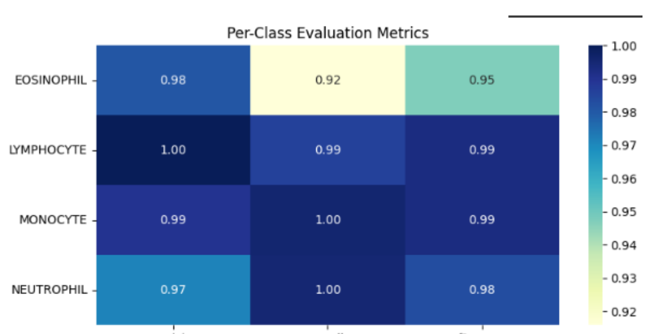

Figure 15: Per-class evaluation for the VGG16 model (precision, recall, F1-score).

C. Precision-Recall and ROC Curves

Precision-Recall and ROC curves are used to evaluate the classification performance of the VGG16 model for each cell category. PR curves highlight the relationship between precision and recall, while ROC curves depict the relationship between the true positive rate and the false positive rate.

The results show very high performance across all categories, with AUC values approaching or reaching 1.00 for both types of curves. This demonstrates the excellent discriminative ability of the VGG16 model among the different white blood cell types.

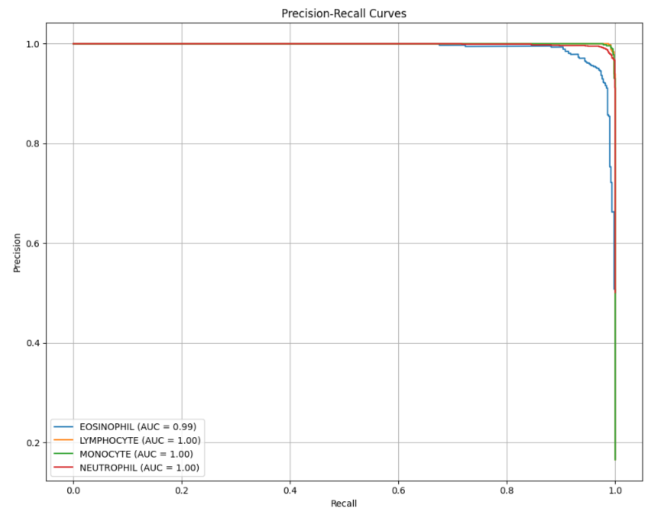

Figure 16: Precision-Recall curves for each cell category of the VGG16 model.

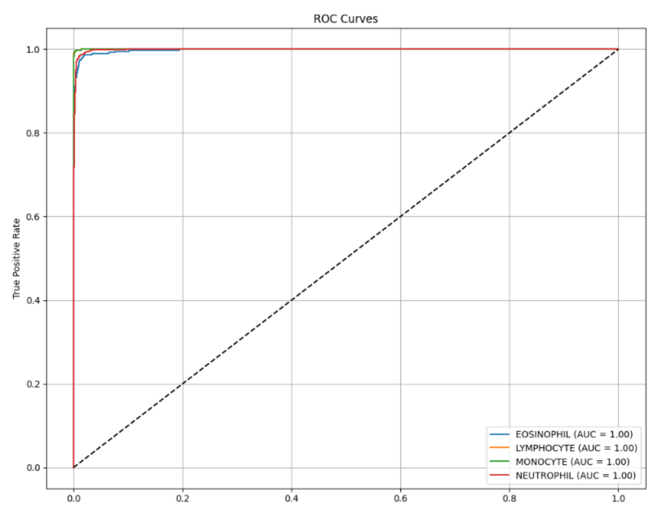

Figure 17: ROC curves for each cell category of the VGG16 model.

D. Explainability with Grad-CAM++

The use of Grad-CAM++ visualizations enables a better understanding of the image regions that contributed most strongly to the final prediction of the model for each cell category:

- **Eosinophil:** Activation is strongly located in the reddish cytoplasmic staining and the nucleus, features that represent key morphological characteristics for eosinophil recognition.

- **Lymphocyte:** The model focuses primarily on the compact, round nucleus, which is strongly stained. This focus suggests an appropriate interpretation of the main discriminative features of lymphocytes.
- **Monocyte:** Warm regions are concentrated on the large, less dense nucleus and the broad cytoplasmic space, confirming that the model takes into account the characteristic size and shape of monocytes.
- **Neutrophil:** The model focuses on the multilobed nucleus, which is the most characteristic morphological feature of neutrophils, demonstrating the ability to recognize even subtle structural differences.

Overall, the visualizations indicate that the VGG16 model relies on morphologically valid features, strengthening the reliability and interpretability of its predictions.

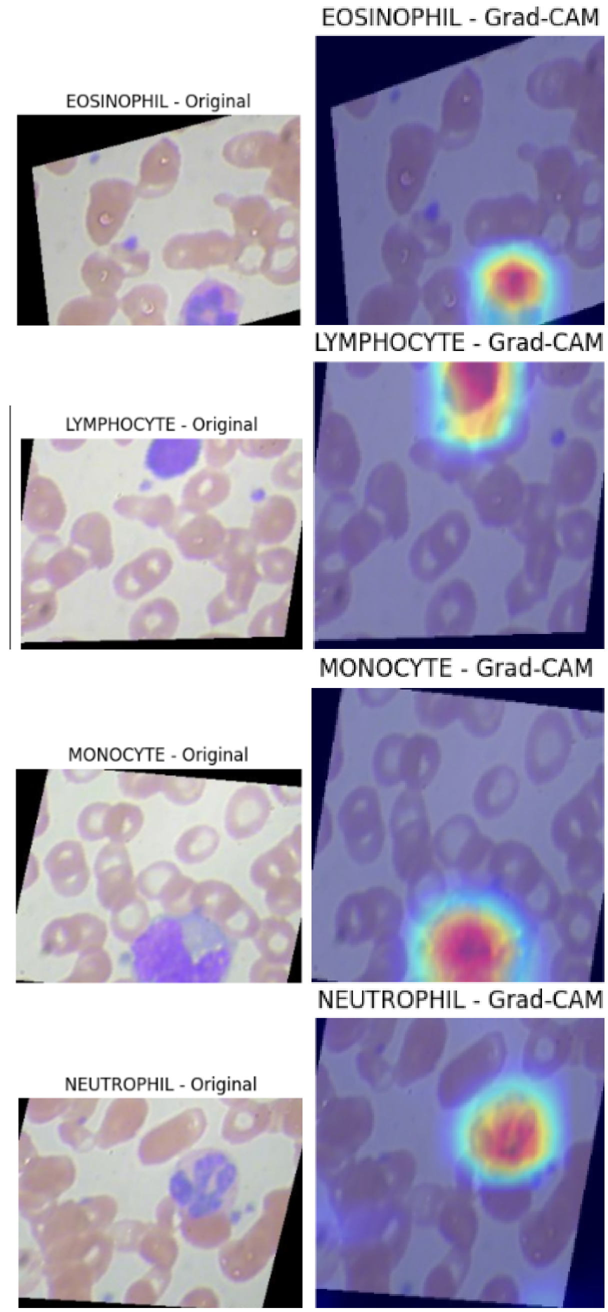

Figure 18: Grad-CAM++ visualizations for each cell category of the VGG16 model. Left: original image. Right: activated regions according to the model.

VI. Results - ResNet18 Model

A. Training Performance

The loss graph (*Loss Curve*) shows a rapid decrease in both training and validation loss during the first epochs, indicating that the model learns effectively from the beginning. As training progresses, both curves converge to low values without signs of overfitting.

Similarly, the validation accuracy curve (*Validation Accuracy*) shows a steady increase and stabilizes above 99%, confirming the model's high generalization performance on unseen data.

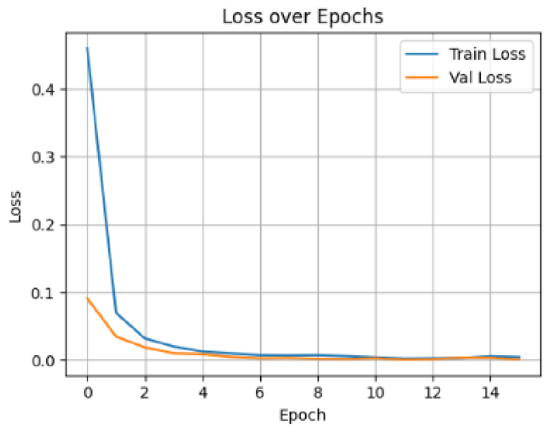

Figure 19: Training and validation loss curve for the ResNet18 model.

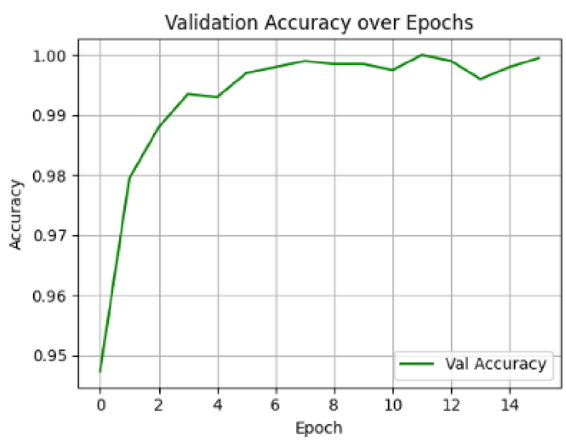

Figure 20: Evolution of validation accuracy for the ResNet18 model.

The ResNet18 model exhibited very rapid convergence, with validation loss approaching zero from the early epochs, while validation accuracy approached 100% after only 6 epochs, indicating excellent generalization capability.

B. Performance Evaluation

The ResNet18 model achieved nearly perfect performance across all categories, with an *accuracy* of 100% and an F1-score equal to 1.00 for every cell type (Figure 21). The confusion matrix shows that errors were minimal, with only 2 incorrect predictions in the *EOSINOPHIL* category, while the remaining classes were classified perfectly.

In addition, both *precision* and *recall* were equal to 1.00 for all classes, indicating that the model produced neither false positive nor false negative results. Model performance is balanced across all classes, as shown by identical values for the *macro average* and *weighted average* metrics, which also reach 1.00.

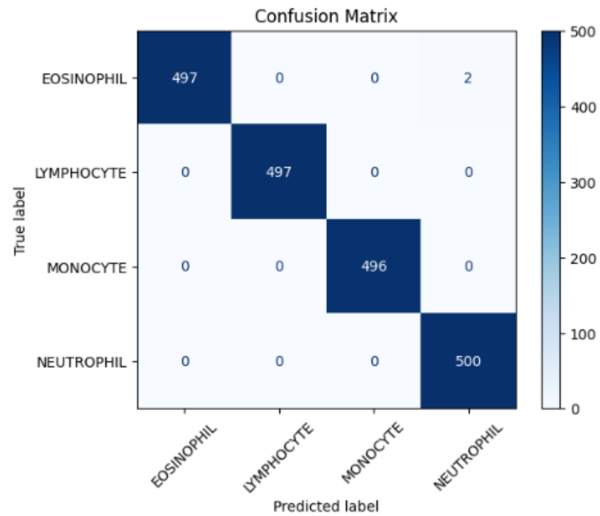

Figure 21: Confusion matrix of the ResNet18 model.

Furthermore, visualization of the feature embeddings using t-SNE (Figure 22) reveals clearly distinct groups for each cell type. This confirms that the network learned discriminative and recognizable feature patterns for each category.

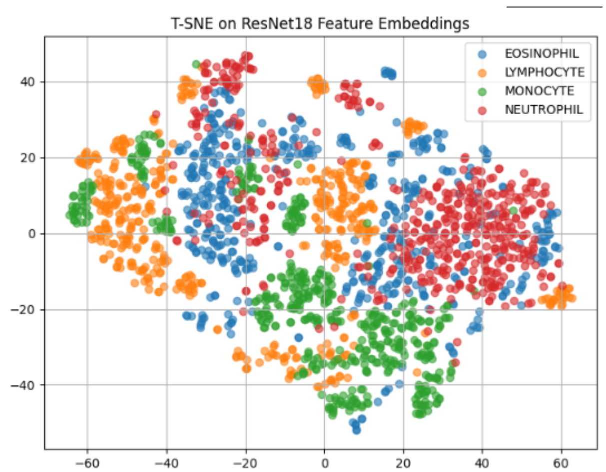

Figure 22: Feature visualization using t-SNE for the ResNet18 model.

The t-SNE clustering confirms the excellent separability of intermediate features for each cell type, providing an additional level of confidence in the model's generalization capability.

C. Precision-Recall and ROC Curves

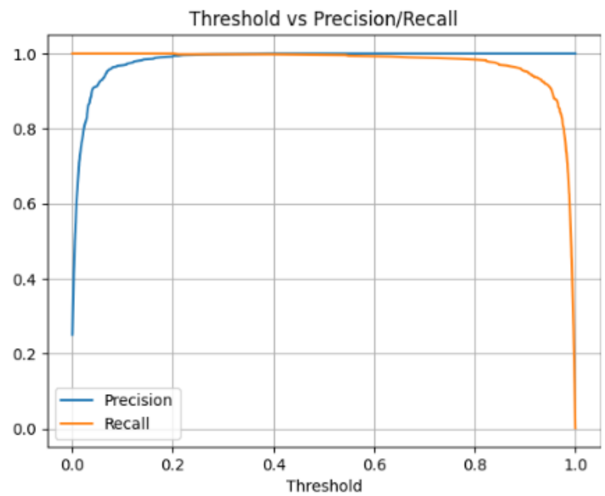

Figure 23: Precision-Recall curves per category for ResNet18.

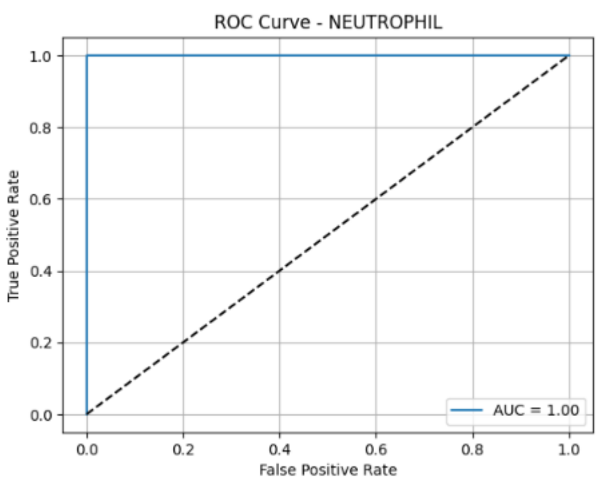

Figure 24: ROC curve for the NEUTROPHIL category (the remaining curves are identical).

All Precision-Recall (PR) and ROC curves demonstrate nearly ideal behavior of the model. The areas under the curves (AUC) are equal to 1.00 for every cell category, indicating perfect discriminative ability among the classes. Specifically:

- **Precision** remains consistently high even at low thresholds.
- **Recall** remains close to 1.00 throughout the threshold range.
- The ROC curves indicate complete separation of positive and negative examples without classification errors.

This level of performance, without any sacrifice in precision or recall, indicates a very well-trained model with no apparent signs of overfitting, which is also supported by the loss and accuracy curves during training.

D. F1-score and Threshold Analysis

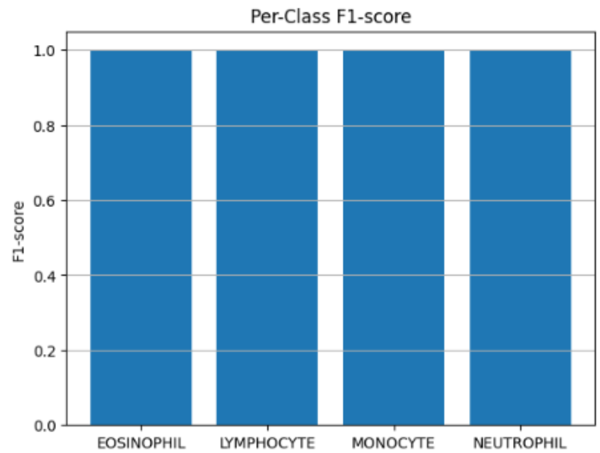

Figure 25: F1-score per category for ResNet18.

All categories achieved an F1-score of 1.00, demonstrating a perfect balance between precision and recall. This performance indicates a model that not only avoids incorrect classifications but also consistently recognizes all samples of each category.

The perfect F1-score for every cell type (EOSINOPHIL, LYMPHOCYTE, MONOCYTE, NEUTROPHIL) demonstrates that the model is equally capable of identifying positive samples and avoiding false positive predictions.

In addition, the behavior of the model with respect to the decision threshold remains stable, without a significant drop in performance across different values. This provides additional reliability and makes the system suitable for application in real clinical scenarios, where stability is critical.

Finally, the analysis of precision and recall as functions of the decision threshold confirms the stable behavior of the model throughout the threshold range, without abrupt declines in either metric.

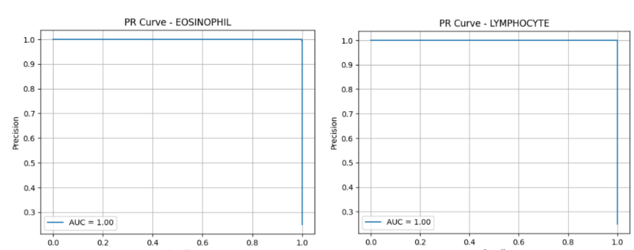

Figure 26: PR curves for EOSINOPHIL (left) and LYMPHOCYTE (right).

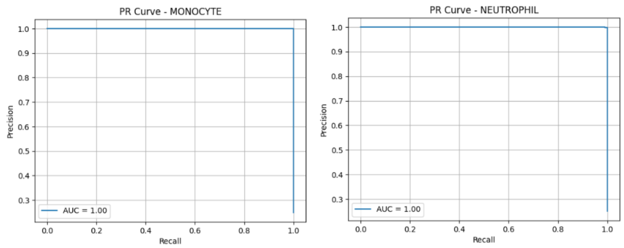

Figure 27: PR curves for MONOCYTE (left) and NEUTROPHIL (right).

E. Explainability with Grad-CAM++

Grad-CAM++ was applied to interpret the decisions of ResNet18 by visualizing the regions on which the network focuses when predicting the category of each cell image.

Figure 28 presents Grad-CAM++ heatmaps for four representative cell images, one for each category. Each pair includes the original image and the corresponding Grad-CAM++ visualization highlighting the model's "attention regions."

- **EOSINOPHIL:** The model focuses on the nucleus, which is multilobed and often contains granules. The warm regions overlap the morphologically distinctive characteristics.
- **LYMPHOCYTE:** The network's attention is concentrated on the round and compact nucleus, with limited peripheral cytoplasmic area. Grad-CAM++ shows strong alignment with the main characteristics of lymphocytes.
- **MONOCYTE:** The focus is distributed across larger regions of the nucleus and the irregular cytoplasmic structure. Distinctive lobulations and cell size are successfully identified by the model.
- **NEUTROPHIL:** The model recognizes the multilobed nucleus and the characteristic cytoplasmic staining. The warm regions are concentrated in the center of the cell, where the main recognizable morphological patterns are located.

The close alignment between the network's attention regions and biologically distinctive characteristics strengthens the interpretability of the model and the confidence of healthcare professionals in its use for clinical applications.

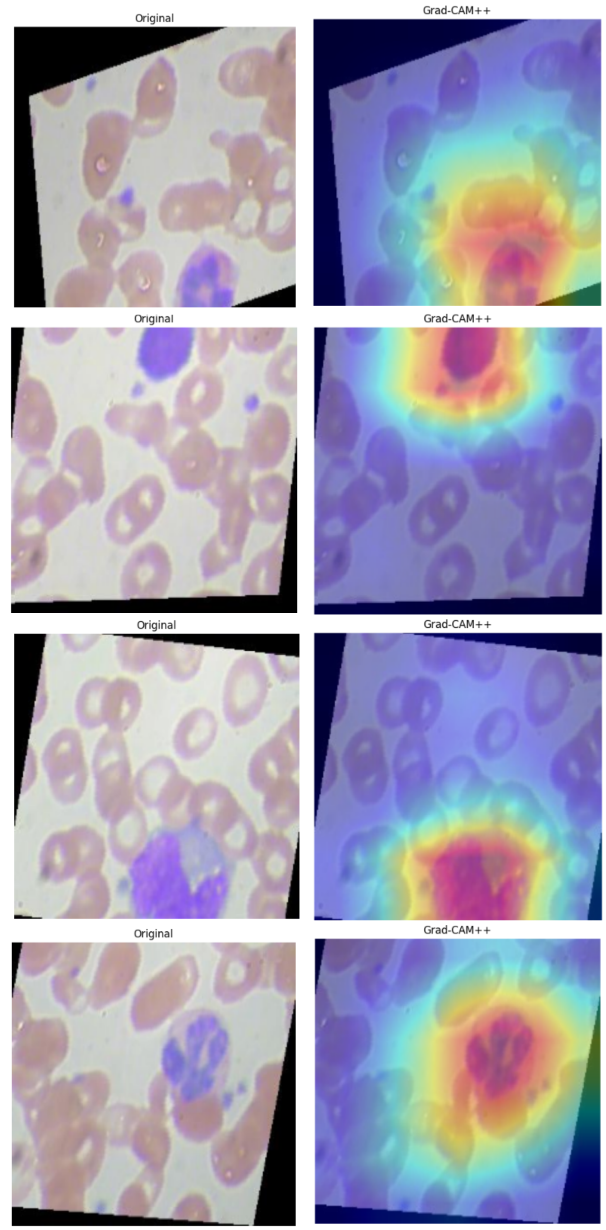

Figure 28: Grad-CAM++ visualizations for each cell type. Left: original image. Right: attention regions of the model.

The Grad-CAM++ visualization confirms that the model focuses on morphologically important regions of the cells when making its classification decisions, strengthening confidence in the interpretability of ResNet18.

F. Comparison of Classification Methods

To evaluate the different architectures used for the cell classification problem, we compared CNN, VGG16, and ResNet18 based on several metrics, including accuracy, precision, recall, and F1-score, both overall and per class. The analysis was performed using confusion matrices, ROC and Precision-Recall curves, reliability diagrams, and Grad-CAM++ visualizations.

The simple CNN demonstrated excellent performance, with an overall accuracy of 99% and an F1-score of 0.99.

Classification was nearly flawless for most categories, with small decreases in recall for the *NEUTROPHIL* class (0.97). The remaining classes were classified with precision and recall values reaching 1.00, indicating strong generalization capability. The ROC and Precision-Recall curves showed AUC = 1.00, while the Grad-CAM++ diagrams demonstrated clear focus on regions of interest within the cells, strengthening interpretability.

The VGG16 architecture produced results similar to the CNN, with an overall accuracy of 96.9% and an F1-score of 0.969. The ROC and PR curves for each class also show AUC = 1.00, while the visualization of embeddings through t-SNE demonstrated satisfactory separability. The use of Grad-CAM++ revealed focused regions within the cells, confirming the interpretability of the model. However, the performance of the model was affected by sporadic errors in cases such as confusion between *NEUTROPHIL* and *EOSINOPHIL*.

ResNet18 achieved the best overall performance, with an accuracy of 99.95% and an F1-score of 1.00. The model classified almost all samples correctly, with zero or very few errors in each category, as shown by the confusion matrix. The ROC and Precision-Recall curves also presented AUC = 1.00 for each class, while the Grad-CAM++ maps accurately localized the relevant regions of interest. Finally, the reliability diagram indicates excellent probability calibration. Overall, ResNet18 represents the most efficient and reliable choice, outperforming the other models in both accuracy and interpretability.

Overall, ResNet18 stood out with the highest accuracy and the most balanced metrics. VGG16 offers competitive performance with lower computational complexity, while the CNN remains a lightweight solution with impressive results, making it suitable for applications where speed and limited computational resources are important.

Table III: Comparative table of classification performance for the three models

Model	Accuracy	Precision	Recall	F1-score
CNN	0.990	0.990	0.990	0.990
VGG16	0.969	0.980	0.969	0.969
ResNet18	0.999	1.000	1.000	1.000

The ResNet18 developed in the present study achieved the highest overall performance, with 99.95% accuracy and an F1-score of 1.00, surpassing the results of previous studies in blood cell classification. The use of Grad-CAM++, reliability diagrams, and data augmentation contributed to the stability, interpretability, and generalization ability of the model. The confusion matrix showed minimal or no classification errors, while ROC and PR curves for each category had AUC = 1.00.

Compared with recent studies in the literature, such as Kumar and Nelson [2], who used EfficientNetB3 and achieved 99.63%, and Asghar et al. [3], who developed a custom CNN with an accuracy of 99.57%, the ResNet18 model in this study reported slightly better performance. The model of Nahzat et al. [4], with an accuracy of 99.5%, is also noteworthy, while the pretrained VGG16 used by Rehman et al. [6] achieved comparable but slightly lower results.

Beyond accuracy, the ResNet18 model demonstrated impressive interpretability and stability across the dataset, making it particularly suitable for clinical applications. Its performance suggests that architectural selection, combined with techniques such as fine-tuning, normalization, and supervised data augmentation, can lead to top-level classification results.

Table IV: Comparison of the ResNet18 performance in the present study with results from the literature

Study	Architecture	Main technique/optimization	Accuracy
Kumar & Nelson (2025) [2]	EfficientNetB3	Fine-tuning, Data Augmentation	99.63%
Asghar et al. (2024) [3]	Custom CNN (VGG+Inception-like)	Dropout, BatchNorm, Data Augmentation	99.57%
Nahzat et al. (2022) [4]	Lightweight CNN	RMSprop optimizer, simple CNN structure	99.50%
Rehman et al. (2018) [6]	VGG16	Transfer Learning, Dropout	-
Biswal et al. (2023) [7]	Inception-ResNet-v2	Fully automated, no manual features	99.00%
Davamani et al. (2024) [9]	VGG16, MobileNet	Balanced dataset, dropout	up to 100% (per class)
Sharma et al. (2022) [10]	DenseNet121	Batch norm, augmentation	98.84%
Girdhar et al. (2022) [5]	CNN	Grayscale conversion, normalization	98.55%
Yıldırım & Cinar (2019) [11]	CNN + image filters	Gaussian/median filters	-
Present study	**ResNet18**	**Grad-CAM++, Reliability Diagram, Augmentation**	**99.95%**

VII. Conclusions

This study focused on the automatic classification of white blood cells (WBCs) from microscopic images using three different deep learning architectures: CNN, VGG16, and ResNet18. The aim was to investigate the performance of these models on the *Blood Cell Images Dataset* and systematically compare them under common experimental conditions using several evaluation metrics.

The analysis of the results showed that all three models achieved high levels of performance, with accuracy above 96%. However, the ResNet18 architecture stood out, achieving 99.95% accuracy and an F1-score equal to 1.00, with nearly perfect classification across all categories. In contrast, the CNN and VGG16 models showed slightly reduced performance in specific classes, such as *Eosinophils* and *Neutrophils*, which was attributed to overlapping morphological characteristics and limited diversity within the dataset.

Model evaluation was performed using confusion matrices, ROC and Precision-Recall curves, as well as Grad-CAM++ maps to study interpretability. ResNet18 achieved AUC = 1.00 for all classes, while the Grad-CAM++ maps showed clear focus on the regions of interest for each cell type, confirming that the model does not rely on random features but on biologically important patterns.

The contribution of careful image preprocessing and data augmentation was particularly important, as these techniques improved model generalization and reduced the risk of overfitting. The use of normalization, dropout, and early stopping also contributed to the stability of the results.

Compared with previous studies in the literature, the performance of ResNet18 exceeds the results of several known approaches. For example, Kumar and Nelson [2] achieved 99.63% accuracy using EfficientNetB3, Asghar et al. [3] achieved 99.57% using a custom CNN, and Nahzat et al. [4] reached 99.5% using a lightweight CNN architecture. Despite the excellent results of these approaches, ResNet18 in the present study not only exceeded these accuracy values but also combined accuracy, interpretability, and simplicity in training.

These findings demonstrate the potential of convolutional networks, and residual networks in particular, for medical image classification. The ability of ResNet18 to learn complex representations without loss of information, due to its skip connections, appears to be critical for successfully distinguishing cell types with high morphological similarity.

For future directions, expansion of the dataset with images from different sources and staining methods is proposed in order to evaluate model generalization on heterogeneous data. In addition, investigation of lightweight models such as MobileNet, or the application of pruning and quantization techniques, could make it possible to integrate these systems into portable devices and microscopes with embedded intelligence.

In conclusion, the present study provides strong evidence that ResNet18 is a highly effective and reliable choice for white blood cell classification, contributing to the direction of automated diagnosis in hematology and opening the way for future clinically applicable solutions.

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